WAFT: The Evolutionary Code Laboratory

SUMMARY

The Physics is the Scint System, which acts as a fitness function through Reality Fracture Detection.

Generated: *2026-01-11 23:06 UTC* | ***Edition:*** *Black & White* | ***Ideas:*** *13*

What is WAFT?

Code as DNA

WAFT is a Python framework for directed evolution of self-modifying AI agents. Think of it as an operating system for AI agent research. Instead of building agents that just execute code, WAFT enables you to breed agents that modify their own code, evolve through mutations, and be tested in fitness systems with complete lineage tracking.

The Three Pillars

The Three Pillars: Substrate, Physics, Flight Recorder

THE SUBSTRATE

Agents write their own Python code. In WAFT, code is DNA. Each agent has a unique genome ID (SHA-256 hash). Agents spawn variants with mutations, evolve by hot-swapping code, and reproduce by creating children with genetic modifications.

THE PHYSICS

The Scint System acts as a fitness function through Reality Fracture Detection, serving as natural selection that eliminates weak mutations. Agents face quests testing error handling: SYNTAX_TEAR (formatting), LOGIC_FRACTURE (math errors), SAFETY_VOID (harmful content), and HALLUCINATION (fabricated facts).

THE FLIGHT RECORDER

A telemetry system generating phylogenetic trees of agent lineage. Every evolutionary action is recorded with complete context: genome ID, parent ID, generation number, event type, payload, and fitness metrics. This enables reconstruction of family trees for scientific publication.

How It Works

WAFT provides project scaffolding through a unified CLI. Run one command to create a fully configured project. Uses uv for package management, creates a _pyrite memory structure, includes CI/CD pipelines, and provides AI agent templates. Everything is file-based with no database, just plain text files that work with git.

Agent Lineage Evolution

QUICK START

Install: `uv tool install waft`
Create: `waft new my_lab`
Verify: `waft verify`

What Makes WAFT Unique: Ambient operation. Self-modifying projects. Meta-framework that orchestrates tools. File-based and git-friendly. Includes gamification, epistemic tracking, and scientific observation with complete lineage tracking.

Key Features

- **Self-Modification:** Agents can modify their own code and evolve
- **Fitness Testing:** Scint Gym evaluates agents through error handling
- **Lineage Tracking:** Complete phylogenetic trees for scientific analysis
- **File-Based:** Everything is plain text, git-friendly, no database
- **Scientific Instrument:** Produces data for research on AI cognition